

Project Design Document

What problem are we solving?

The Division Manager of Urban Forestry wants to understand which work zones and contractors are driving overall SLA compliance below target, and is the bottleneck caused by backlog age in specific service types or caused by low workforce capacity?

How will we analyze the problem?

Perform root cause analysis of tree maintenance work orders from 2025 to answer the following business questions:

- Why is overall SLA compliance dropping below target?
- Why are tree trimming jobs missing their SLA at a high rate?
- Which work zone is most underperforming on tree trimming? Why?

What is the project's scope?

1. SLA Breach Rate vs Tree Removals

- Review SLA breach rates for tree removal work orders generated across all zones. Highlight the average volume of work orders generated and the average breach rate. Is the relationship between volume and breach rate positively or negatively correlated?
 - Positive correlation is expected – SLA breach rate increases as volume increases. Explore any negative correlations.
 - Use line chart with dual axis

2. SLA Breach Rate vs Tree Trimming

- Review SLA breach rates for tree trimming work orders generated across all zones. Highlight the average volume of work orders generated and the average breach rate. Is the relationship between volume and breach rate positively or negatively correlated?
 - Positive correlation is expected – SLA breach rate increases as volume increases. Explore any negative correlations.
 - Use line chart with dual axis

3. Work Orders Completed Per Day Per Crew

- Review the number of work orders completed per day per crew across all contractors to see if crew productivity is lacking in the zones with the highest demand.
 - Use horizontal bar chart

4. Average Lead Time

- Review the average lead time, from intake to completion, for each contractor. What stands out? What conclusion(s) can we draw from this information when combined with what we discover from sections 1 through 3 above.
 - Use horizontal bar chart

What is the dashboard design?

The dashboard will include visualizations for metrics leadership can use to quickly spot developing backlogs by looking at work orders closed versus work orders created each month, display backlogs by zone to see when resource reallocation maybe necessary, display work orders closed by contractor crews versus in-house crews to compare productivity levels, and display the volume of work orders by priority level to highlight any growing backlog of emergency requests.

Include BANs for critical KPIs (Tree Maintenance Cost, Open Backlog, Emergency Work Orders, Average Lead Time, SLA Breach Rate across all work types).

Include global filters to allow filtering by Work Type, by Month, and by Zone.